جـامـعـة الـشـرق الأوسط
MIDDLE EAST UNIVERSITY
Amman - Jordan

Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	1st	**Academic Year**	2025/2026

Course Details

Course Title	Software Design			**Course No.**	0442502	
Type of Learning	☐On- campus ■Blended (................Platform) ☐ Online (..............Platform)					
Credit Hours	3	**Theoretical**	3	**Practical**	0	
Course Level (according to JNQF)			7	**JNQF Hours***	12.3	
Pre-requisite	Software Engineering - 0442401			**Co-requisite**	N/A	

Course Instructor Information

Instructor Name	Dr. Hani Barjes Al-bloush		
Instructor E-mail	halbloush@meu.edu.jo		
Course Time	09:00 – 10:00 (Mon., Wed.,)		
Office Hours	TBA		
Office Number	N-144	**Office Phone**	N/A
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Modern Systems Analysis and Design, J. S. Valacich, Joey F.George and Jeffrey A. Hoffer, Published by Pearson, 9th Edition, September 18th 2020

2. References

- System Analysis & Design in a Changing World by John W. Satzinger, Robert B. Jackson and Stephen D. Burd, 7th Edition, 2016. ISBN 978-1-305-11720-4.
- Systems Analysis and Design, 11th Edition, Scott Tilley and Harry J. Rosenblatt, 2017, Print ISBN: 9781305494602

Teaching Methods

1. Lectures/ Discussion
2. Case Studies
3. Assignments
4. Guest Speakers
5. Projects/ Presentations

Course Description

This course offers a comprehensive understanding of object-oriented principles and UML for software modeling. Through lectures and practical exercises, students will learn to effectively apply object-oriented concepts to design software systems using UML diagrams that represent structure, behavior, and interactions. They will gain hands-on experience with UML modeling tools and explore database design using Entity-Relationship Diagrams (ERDs). Additionally, the course covers user interface design principles, enabling students to create user-friendly interfaces. Finally, students will work in teams on a real-life project, integrating their skills in object-oriented design and modeling to complete the design phase successfully.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLO's	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Identify the importance of software modeling and Unified Modeling Language (UML).	PLO1	K1-I		
2	Investigate various UML diagrams, including Use Case, Activity, Sequence, and Class diagrams to visualize software system	PLO1	K1-P		
3	Identify the database requirements and design schemas using ERDs to ensure data integrity, efficiency, and scalability.	PLO1	K1-I		
4	Identify user interface principles to design interfaces, dialogs, forms, and reports that meet user needs and user satisfaction.	PLO2		S1-I	
5	Execute the knowledge and skills of software design to complete the design phase of a real-life project.	PLO2		S1-P	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	**Introduction to Software Modeling and UML:** - What is modelling? - What is UML? - Modelling examples and general concepts	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
2&3	**Use Case diagram:** -What is Use Case? -Definitions, Symbols & Relations -Drawing Use Case diagram -Writing Use Case descriptions	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Blended	2	Interactive content & Quiz	Summative
	Activity Diagram: -What is an activity diagram?	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative

3&4	-Definition and symbols -flow and decisions -Drawing Activity diagram	Lecture (2)	Lecture On- campus	2	Hands-on Activity Assignment	Summative
		Lecture (3)	Blended	2	Interactive content & Quiz	Summative
5	**System & Detailed Sequence Diagram:** -Introduction to sequence diagram -Sequence diagram elements -Sequence diagram as full model	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Hands-on Activity Assignment	Summative
		Lecture (3)	Blended	2	Interactive content & Quiz	Summative
6 & 7	**Domain & Detailed Class Diagram:** -Object and Classes -association and type of association (Multiplicity) -Aggregation -Composition -Inheritance -Generalization	Lecture (1)	Lecture On- campus	2	Quiz1	Summative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Blended	2	Interactive content & Quiz	Summative
8	**Midterm Exam**	-	Assessment	1 & 2	Questions & Rubric	Summative
9	**Entity Relationship diagram:** -Entities -Attributes -Candidate Keys and Identifiers -Relationships **Designing databases:** -The process of database design	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Blended	3	Interactive content & Quiz	Summative
10	**Designing Forms and Reports:** -The process of designing forms and reports -Deliverables and outcomes	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Hands-on Activity Assignment	Summative
		Lecture (3)	Blended	4	Interactive content & Quiz	Summative
11	**Formatting Forms and Reports:** -General Formatting Guidelines -Highlighting Information -Color versus No Color -Displaying Text -Designing Tables and Lists **Assessing Usability:** -Usability Success Factors -Measures of Usability	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Blended	4	Interactive content & Quiz	Summative
12	**Design Interface and Dialogues:** -The process of designing forms and reports -Deliverables and outcomes -Interaction methods and device	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Blended	4	Interactive content & Quiz	Summative
	-Designing Interfaces layouts -Structuring data entry	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative

13	-Controlling data input -Providing feedback -Providing Help -Designing the Dialogue Sequence	Lecture (2)	Lecture On- campus	4	Hands-on Activity Assignment	Summative
		Lecture (3)	Blended	4	Interactive content & Quiz	Summative
14	**Design Interface & dialogues:** -Building Prototypes and Assessing Usability -Designing Interfaces and Dialogues in Graphical Environments	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Blended	4	Interactive content & Quiz	Summative
15	**Project**	-	Presentation	5	Project	Summative
16	**Final Exam**	-	Assessment	1,2,3,4 & 5	Questions & Rubric	Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Midterm Exam	25%	Week 8
Assignment	15%	Week 3, 5, 10, 13
Quiz	5%	Week 6
Project	15%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	▪ No makeup quizzes. ▪ A minimum of 1 quiz is given. ▪ Each Quiz is out of 10. If more quizzes are given, then the lowest quiz's grade is dropped.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	The last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exams or quizzes is an illegal and unethical activity. ▪ Standard MEU policy will be applied. All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. ▪ Sign-in sheets will be circulated. If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.

Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	▪ Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Assignments are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application.
	More details about phases and topics of the projects will be given later on.
Others	

Rubrics of Course Project

Criteria	3 Marks	2 Marks	1 Mark
Identification of Project Challenges & Research Gap	Clearly identifies significant challenges and articulates a well-defined research gap.	Identifies relevant challenges and research gap with some clarity issues.	Vague or poorly defined challenges and research gap.
Research Objectives and Methodology	Formulates clear research objectives and selects an appropriate methodology.	Objectives are mostly clear; methodology selection is appropriate but lacks depth.	Unclear objectives; methodology choice is poorly justified
UML Models and ERD	Produces complete UML models and ERD that are accurate and well-structured.	UML models and ERD are mostly complete but may have some inaccuracies.	Incomplete or poorly designed UML models and ERD.
Prototype Development	Prototype is well-designed using Figma, showcasing creativity and functionality.	Prototype is functional with some design issues.	Prototype is poorly designed or lacks functionality.
Presentation and Q&A	Presentation is clear, engaging, and well-organized; confidently answers questions.	Presentation is mostly clear; answers questions adequately but lacks depth.	Presentation lacks clarity; struggles to answer questions.

* **JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3 hours	14	42

Tutorials	0	0	0
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1 hours	1	1
Second Exam (If applicable)	0	0	0
Final Exam	2 hours	1	2
Quizzes	0.5 hour	2	1
Self-Learning Activities			
Assignments/ Homework	3 hours	6	18
Case Studies	0	0	0
Presentations	1.5 hours	2	3
E-learning	0	0	0
Extra Readings	4	14	56
Total Hours	**123**		
JNQF Hours	**123/10 = 12.3**		